ITUMELENG NTELA

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PROFESSIONAL SUMMARY

BSc Information Technology degree holder pursuing Honours in Computer Science and Information Technology at North-West University. Experienced in full-stack development, cloud computing, artificial intelligence, and data engineering. Proven track record of taking full ownership of complex technical systems from design through to deployment. Seeking a graduate software engineering program where I can apply strong analytical thinking and problem-solving ability to deliver impact from day one.

TECHNICAL SKILLS

Programming	Python, Java, C#, C++, JavaScript, SQL
AI and Machine Learning	Reinforcement Learning, Q-Learning, Machine Learning, Microsoft AI Fluency
Cloud and AWS	AWS Fundamentals, Google Cloud Platform, Microsoft Azure
Data Engineering	Apache Kafka, Power BI, Data Analytics, ETL Pipelines, NoSQL
Software Development	OOP, Data Structures, Algorithms, Git, Agile, CI/CD, SDLC, System Design
Web and Full-Stack	HTML, CSS, JavaScript, Node.js, React, REST APIs, JSON
Databases	SQL Server, MongoDB, Azure Cosmos DB, NoSQL, Data Modelling
Frameworks	.NET Framework, WinForms, Visual Studio, VS Code, PyCharm, Git

PROJECTS

Cloud-Based Oncology Management System | *2026 Honours Research Project*

- Designing and developing a cloud-hosted system to digitise patient records in an oncology setting, reducing reliance on paper-based processes.
- Conducting requirements analysis to identify workflow inefficiencies and translate them into system specifications.
- **Technologies**: Google Cloud, System Analysis & Design, Database Design.

Data Warehouse & Analytics | *2026 Independent Project*

- Designed and implemented a full data warehousing solution including Extract, Transform and Load pipelines using SQL Server Integration Services (SSIS)
- Technologies: SQL Server, SSRS, SSAS, SSIS

Smart City Unified Threat Intelligence Platform| *2026 Independent Project*

- Designed and built a real-time multi-stream threat detection system processing an Azure Event Hub stream.
- Implemented Azure Stream Analytics with event-time windowing.
- Stored enriched event data in Azure Cosmos DB and built Power BI dashboards visualising KPIs.
- **Technologies**: Microsoft Azure, Event Hubs, Stream Analytics, Cosmos DB, Power BI, Python, SQL, IoT

Reinforcement Learning AI Agent | *2026 Independent Project*

- Built a Q-Learning AI agent for the traditional African game Mancala to achieve an 85% win rate.
- Engineered a shaped reward function to accelerate convergence and avoid sparse reward problems
- Developed a fully playable web interface supporting Human vs AI, AI vs AI, and hint modes, bringing the trained agent to life in a browser.
- **Technologies**: Python, Q-Learning, Reinforcement Learning.

StudentDev Hub Platform | *2025 Group Leader*

- Led an Agile team to deliver a full-stack web application using Node.js, React and MongoDB, managing sprints, code reviews, and final delivery.
- **Technologies**: Node.js, React, MongoDB, REST APIs, Agile.

ClearVue BI Platform | *2025 Team Member*

- Built a real-time Business Intelligence platform processing live data streams with Apache Kafka.
- Implemented MongoDB for scalable NoSQL data storage and designed Power BI dashboards for executive-level decision-making.
- **Technologies**: Apache Kafka, MongoDB, Power BI, Node.js.

EDUCATION

BSc Honours in Computer Science and Information Technology | (In Progress)
North-West University

- Research Project: Cloud-Based Oncology Management System on Google Cloud
- Key Modules: Data Warehouses, Databases, Information Systems Engineering, Artificial Intelligence

BSc in Information Technology | 2023 - 2025 (Completed)
North-West University

- Golden Key International Honour Society - Top 15%

WORK EXPERIENCE

Peer-Assisted Learning Facilitator - Java (CMPG211) | North-West University · *2025*

- Facilitated structured weekly sessions on object-oriented programming and Java fundamentals for groups of students.
- Developed session plans and adapted delivery based on student feedback, demonstrating communication and leadership skills.

Student Learning Assistant - Python (CMPG111) | North-West University · *2025*

- Supported students with Python fundamentals, debugging techniques, and computational thinking.
- Delivered one-on-one and group mentoring sessions to improve academic performance.

Student Learning Assistant - C++ (CMPG121) | North-West University · *2024*

- Assisted students with C++ syntax, memory management, and problem-solving strategies.
- Conducted individual consultations and group workshops to reinforce lecture content.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

2026 Vodacom Digital Skills Programme - Short Course
2025 Microsoft AI Fluency Certificate
2025 FNB App Academy - Full Stack Development
2024 Chongqing University of Technology - Cyber Security Training

REFERENCES

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